

Problem 1: Unique Factorization

(a)

In the ring $\mathbb{Z}[i]$, factor the element $17 + 9i$ into a product of irreducibles.

Proof.

$$N(17 + 9i) = 2 \cdot 5 \cdot 37$$

$$\frac{(17 + 9i)(1 - i)}{(1 + i)(1 - i)} = 13 - 4i$$

$$\frac{(13 - 4i)(2 + i)}{(2 + i)(2 - i)} = 6 + i$$

$$17 + 9i = (1 + i)(2 - i)(6 + i)$$

These are irreducible since their norms are primes. ■

(b)

In the ring $\mathbb{Z}[\omega]$, factor the elements $17 + 9\omega$ and $17 - 9\omega$ into products of irreducibles.

Proof.

$$N(17 - 9\omega) = 523 \text{ which is prime}$$

Thus, $17 - 9\omega$ is irreducible.

$$N(17 + 9\omega) = 7 \cdot 31$$

$$\frac{(17 + 9\omega)(3 + \omega)}{(3 + \omega)(3 + \omega^2)} = (6 + 5\omega)$$

$$17 + 9\omega = (3 + \omega^2)(6 + 5\omega)$$

These are irreducible since their norms are primes. ■

Problem 2: Ideal Factorization

(f)

Show that $(7) = \mathfrak{p}_3 \mathfrak{p}_4$ for some maximal (hence prime ideals) $\mathfrak{p}_3$ and $\mathfrak{p}_4$. Write down what $\mathfrak{p}_3$ and $\mathfrak{p}_4$ are explicitly, i.e., identify generators for them.

Proof.

First we know that $3^2 = 9 \equiv -5 \pmod{7}$.

Thus, consider $(7, 3 - \sqrt{-5})(7, 3 + \sqrt{-5}) = (49, 21 - 7\sqrt{-5}, 21 + 7\sqrt{-5}, 14) = (7)$.

Since $49 - 3 \cdot 14 = 7$, and the rest are divisible by 7.

Want to show that $(R/(7, 3 - \sqrt{-5})) \cong \mathbb{Z}/7\mathbb{Z}$

Define: $\phi : R \rightarrow \mathbb{Z}/7\mathbb{Z}$ s.t. $\phi(a + b\sqrt{-5}) = a + 3b$.

Homomorphism:

$$\phi(a + b\sqrt{-5} + c + d\sqrt{-5}) = a + c + 3(b + d) = \phi(a + b\sqrt{-5}) + \phi(c + d\sqrt{-5})$$

$$\begin{aligned} \phi((a + b\sqrt{-5})(c + d\sqrt{-5})) &= \phi(ac + (ad + cb)\sqrt{-5} - 5bd) = ac - 5bd + 3(ad + cb) = \\ &= ac + 9bd + 3(ad + cb) = (a + 3b)(c + 3d) = \phi(a + b\sqrt{-5})\phi(c + d\sqrt{-5}) \end{aligned}$$

$$\phi(1) = 1$$

Surjective:

$$\text{Let } x \in \mathbb{Z}/7\mathbb{Z}. \phi(x) = x$$

Kernel:

$$\text{Let } x = a + b\sqrt{-5} \in \ker \phi. \phi(x) = 0 = a + 3b$$

$$\text{Thus, } a = -3b + 7k \text{ for some } k \in \mathbb{Z}.$$

$$x = -3b + 7k + b\sqrt{-5} = 7k + b(-3 + \sqrt{-5}) \in (7, -3 + \sqrt{-5}) = (7, 3 - \sqrt{-5}).$$

$$\text{Let } x \in (7, 3 - \sqrt{-5}). \text{ Thus, } x = 7a + b(3 - \sqrt{-5})$$

$$\phi(x) = \phi(7a) + \phi(3 - \sqrt{-5}) = 0 + 3 - 3 = 0.$$

$$\text{Thus, } R/(7, 3 - \sqrt{-5}) \cong \mathbb{Z}/7\mathbb{Z}.$$

Thus, $(7, 3 - \sqrt{-5})$ is a maximal ideal. Do something similar for $(7, 3 + \sqrt{-5})$ ■

(g)

Show that $\mathfrak{p}_3$ and $\mathfrak{p}_4$ are not principal, but their squares are principal.

Proof.

Assume they are principal then there is an $a \in \mathbb{Z}[\sqrt{-5}]$ s.t. $(a) = (7, 3 - \sqrt{-5})$.

Then $N(a) \mid 49$ and $N(a) \mid 14$. Thus, $N(a) \mid 7$. Thus, $N(a) = 7$. Since $(7, 3 - \sqrt{-5})$ is proper.

$$\text{Thus, } 7 = c^2 + 5d^2.$$

Consider $7 \equiv c^2 + 5d^2 \pmod{5}$.

Thus, $7 \equiv c^2 \pmod{5}$.

Thus, $2 \equiv c^2 \pmod{5}$. There is no solution.

Thus, there is no integer solution to this equation. Thus, a does not exist.

Do a similar thing for the other ideal. Thus, we have show that $(7, 3 - \sqrt{-5}), (7, 3 + \sqrt{-5})$ are not principal.

$$(7, 3 - \sqrt{-5})(7, 3 - \sqrt{-5}) = (49, 7(3 - \sqrt{-5}), 4 - 6\sqrt{-5})$$

Consider $(2 - 3\sqrt{-5})$

$$49 = (2 + 3\sqrt{-5})(2 - 3\sqrt{-5})$$

$$4 - 6\sqrt{-5} = 2(2 - 3\sqrt{-5})$$

$$7(3 - \sqrt{-5}) = (3 + \sqrt{-5})(2 - 3\sqrt{-5})$$

$$2 - 3\sqrt{-5} = 49 - 3(21 - 7\sqrt{-5}) + 4(4 - 6\sqrt{-5})$$

$$\text{Thus, } (2 - 3\sqrt{-5}) = (7, 3 - \sqrt{-5})^2$$

The same works for $(2 + 3\sqrt{-5}) = (7, 3 + \sqrt{-5})^2$ since just conjugate all the equations above. ■

(h)

Write a computer program to determine which of the first hundred odd primes remain prime in R . Do you notice a pattern? (Please find a partner to work with on this problem. Indicate in your answer the student that you worked with.)

Proof.

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[11, 13, 17, 19, 31, 37, 53, 59, 71, 73, 79, 97, 113, 131, 137, 139, 151, 157, 173, 179, 191, 193, 197, 199, 211, 233, 239, 251, 257, 271, 277, 293, 311, 313, 317, 331, 337, 353, 359, 373, 379, 397, 419, 431, 433, 439, 457, 479, 491, 499]

The pattern seems to be $11, 13, 17, 19 \pmod{20}$ ■

Problem 3: Euclidean domains

Show that $\mathbb{Z}[\sqrt{-2}]$ is a Euclidean domain.

Proof.

Let the norm function as our size function.

Let $x = a + b\sqrt{-2}, y = c + d\sqrt{-2} \in \mathbb{Z}[\sqrt{-2}]$ s.t. $y \neq 0$.

Consider $x/y = e + f\sqrt{-2}$ s.t. $e, f \in \mathbb{Q}[\sqrt{-2}]$.

Let $g, h \in \mathbb{Z}$ s.t. g has a gap of less than or equal to one half away from e and same for h and f .

Thus,

$$\begin{aligned} a + b\sqrt{-2} &= (c + d\sqrt{-2})(e + f\sqrt{-2}) \\ &= (c + d\sqrt{-2})(g + h\sqrt{-2} + (e - g) + (f - h)\sqrt{-2}) \\ &= (c + d\sqrt{-2})(g + h\sqrt{-2}) + (c + d\sqrt{-2})((e - g) + (f - h)\sqrt{-2}) \end{aligned}$$

$$N((c + d\sqrt{-2})((e - g) + (f - h)\sqrt{-2})) = N((c + d\sqrt{-2}))N((e - g) + (f - h)\sqrt{-2})$$

$$N((e - g) + (f - h)\sqrt{-2}) = (e - g)^2 + 2(f - h)^2 \leq 1/4 + 1/2 = 3/4$$

$$\text{Thus, } N(r) \leq 3/4 N(c + d\sqrt{-2})$$

Thus, $\mathbb{Z}[\sqrt{-2}]$ is a Euclidean domain. ■